Ziyuan Zhao

Curriculum vitae

School of Mathematical Sciences, University of Science and Technology of China
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Education

Present	Ph.D. student in Mathematics University of Science and Technology of China Advisor: Jie Ma
Sep. 2021 - Jun. 2024	M.S. in Statistics, University of Chinese Academy of Sciences Academy of Mathematics and Systems Science, Chinese Academy of Sciences
Sep. 2017 - Jun. 2021	B.S. in Statistics School of Mathematical Sciences, Peking University

Research interests

Combinatorics, particularly extremal and structural graph theory. My current work has a particular focus on cycles and paths, graph connectivity, and vertex-transitive graphs.

Papers and preprints

7. Linear circumference in vertex-transitive graphs
With Jie Ma.
Preprint, 2026. [arXiv:2610.02053]

6. Longest cycles intersect linearly in highly connected graphs
With Jie Ma and Bo Ning.
Submitted. [arXiv:2609.20724]

5. An exact formula for Erdős' problem 1005
With Yanmohan Wang and Mingxu Xie.
Preprint, 2026. [arXiv:2608.15681]

4. Longest cycles and Dirac-type results in highly connected graphs
With Jie Ma and Bo Ning.
International Mathematics Research Notices, to appear. [arXiv:2606.03696]

3. Dean's conjecture and cycles modulo k
With Yufan Luo and Jie Ma.
Submitted. [arXiv:2601.13552]

2. Intersections of longest cycles in vertex-transitive and highly connected graphs
With Jie Ma.
Submitted. [arXiv:2508.17438]

1. Leaf-to-leaf paths and cycles in degree-critical graphs
With Francesco Di Braccio, Kyriakos Katsamaktsis, Jie Ma and Alexandru Malekshahian.
Combinatorica 46, Article 11 (2026). [arXiv:2504.11656] [journal]

Talks

18 Sep. 2026
Longest cycles: length, intersections and transversals
Combinatorics Seminar, Duke University (online)
[slides]

5 Aug. 2026
Longest cycles meet linearly in highly connected graphs
Innovations and Challenges in Extremal Combinatorics
Institute for Mathematical Sciences, National University of Singapore
[slides]

22 Jul. 2026
Longest cycles meet linearly in highly connected graphs
12th National Conference on Combinatorics and Graph Theory
Zibo, China
[slides]

4 Dec. 2025
Intersections of longest cycles in vertex-transitive and highly connected graphs
PhD Workshop on Graph Theory and Combinatorics
Fudan University, Shanghai, China
[slides]

13 Jun. 2025
Leaf-to-leaf paths and cycles in degree-critical graphs
Frontiers in Combinatorics and Theoretical Computer Science (BIRS-IASM)
Hangzhou, China
[slides]